

QueryNest Multimodal RAG - Sample Document

QueryNest is a multimodal document intelligence system. It ingests complex documents such as PDF, images, tables and equations, then enables hybrid retrieval and citation-based answering. The core pipeline is: parse, index, hybrid retrieval, rerank, context construction, and final answer generation.

Hybrid retrieval in QueryNest combines Dense vector retrieval, BM25 keyword retrieval, and LightRAG knowledge-graph retrieval. All candidate hits are fused with Reciprocal Rank Fusion (RRF), deduplicated, and finally reranked before the context window is assembled.

Method	Precision	Recall
Dense only	0.62	0.55
BM25 only	0.48	0.40
Hybrid (RRF)	0.78	0.71

